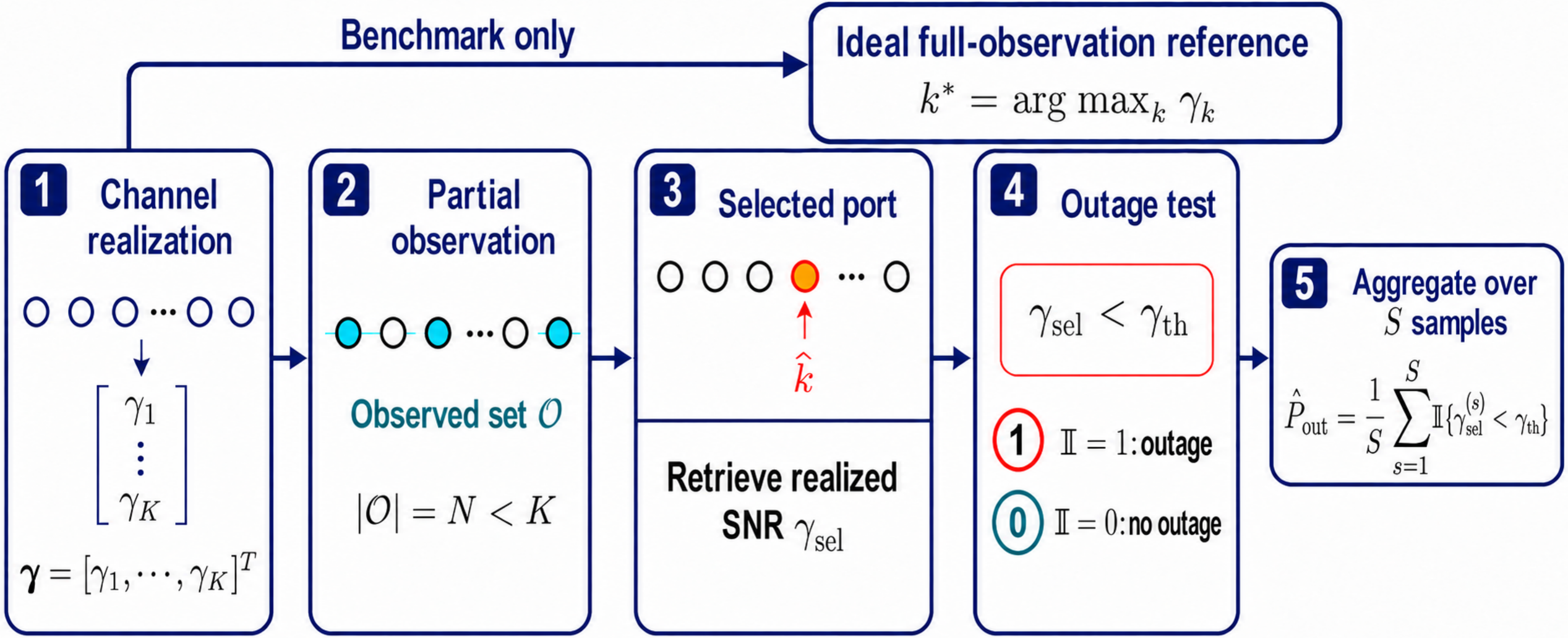

Benchmark only
Ideal full-observation reference
$k^* = \arg\max_k \gamma_k$
1
Channel realization
$\boldsymbol{\gamma} = [\gamma_1, \cdots, \gamma_K]^T$
2
Partial observation
Observed set $\mathcal{O}$
$|\mathcal{O}| = N < K$
3
Selected port
$\hat{k}$
Retrieve realized SNR γ_{sel}
4
Outage test
$\gamma_{\mathrm{sel}} < \gamma_{\mathrm{th}}$
1 $\mathbb{I} = 1$: outage
0 $\mathbb{I} = 0$: no outage
5
Aggregate over S samples
$\hat{P}_{\mathrm{out}} = \frac{1}{S} \sum_{s=1}^{S} \mathbb{I}\{\gamma_{\mathrm{sel}}^{(s)} < \gamma_{\mathrm{th}}\}$